Behavior:

The mouse cursor acts as the sun position and the vehicle will move towards it with a speed (minimum 0, maximum 4) depending on its distance from the sun (minimum 0, maximum 500) .

i.e. when distance is approaching 0, speed will approach to 4 and vice versa.

Logic:

I have tried to explain every step in the code file. Some of the important parts are:

1. Defining a turn rate: it is to ensure that vehicle and sensor isn't turning too fast because it can miss the sun and won't display a smooth turning movement.
2. I used min and max functions to make sure that correct values are chosen in case of negative direction or magnitude. (both for direction and speed of the vehicle)
3. An arbitrary speed and distance was defined to calculate the intensity which will control speed of the vehicle.

$$intensity = max(0, max_distance - distance) / max_distance$$

Settings:

Sun: radius=30, color=yellow, position=(600, 300)

Vehicle: radius=50, color=red, position=(300, 500)

Sensor: radius=15, color=green, position=45 degree angle to vehicle

Maximum distance between vehicle and sun= 500

Maximum speed= 4

Turn rate= 2 degrees per frame

Sun position= cursor position